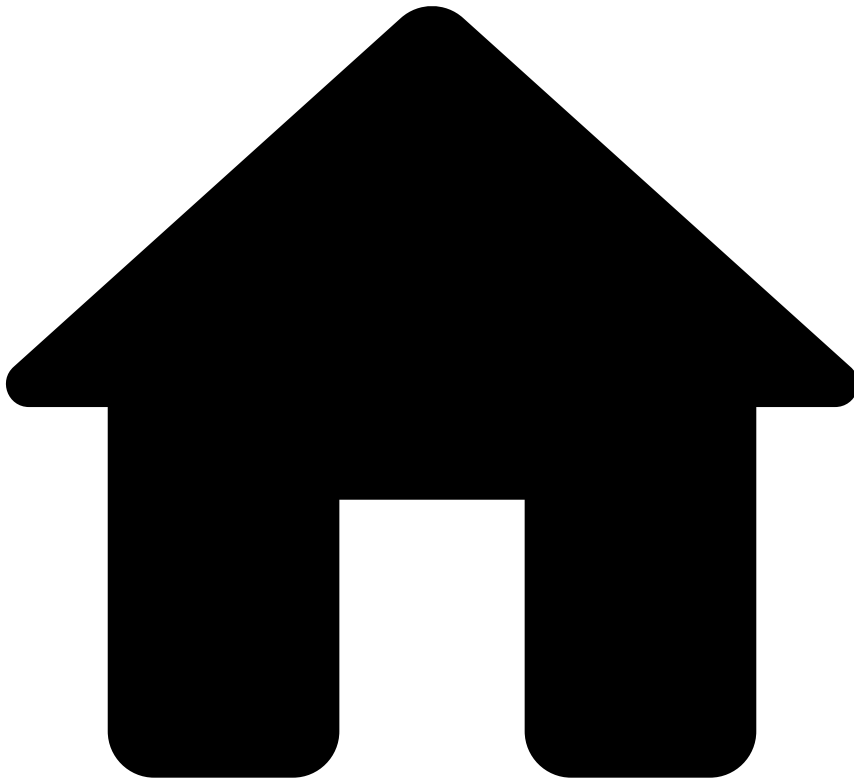


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- [References](#)
- [Rules Projects](#)
- Cards - Fraud Rules

On this page

Cards - Fraud Rules

Was this page helpful? 

Description

Contains rules that look for standard fraud patterns. If triggered, most rules included in this rules project do not auto decline a card. Instead, they mark the event as an alert which can then be further investigated by fraud analysts in Case Manager. Examples of typical patterns covered by these rules include:

- High-amount spending and increased number of transactions in a short period of time.
- High-amount spending in the same merchant in a given time frame by card not present.
- Repeated transactions with the same amount.
- Number of transactions above a certain threshold.

- Unexpected spending behavior.
- High cumulative spending amounts.
- Transactions in unusual cities and/or currencies.
- Multiple CVV attempts in a short period of time.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Cards-Velocity-1

Comment

Decline transaction and flag card to be limited if travel velocity between the last transaction within the past 1 hour and the current transaction is unreasonably high. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Velocity-1]
- Limit/Review Card
- Alert

Variables

```
// maximum velocity in km/h
VELOCITY_LIMIT = cards__rules_thresholds["max_cp_card_term_vel_1h"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
event.card_presence_flag ?? false
and
// Limits
event.cp_card_term_vel_1h >= VELOCITY_LIMIT
```

Explanation

[Cards-Velocity-1] - Decline - Card travel velocity ({event.cp_card_term_vel_1h} km/h) between last and current CP transactions in last 1h is higher than limit ({localvar.VELOCITY_LIMIT} km/h).
'{eventname.card_id}' {event.card_id} flagged to be limited.

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-2

Comment

Decline transaction and flag card to be limited if number of transactions made by card within the past 24 hours is above a set threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Velocity-2]
- Limit/Review Card
- Alert

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_count_24h"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.auth_cardid_count_1d >= COUNT_LIMIT
```

Explanation

[Cards-Velocity-2] - Decline - Card transactions ({event.auth_cardid_count_1d}) in the last 24 hours are above the limit ({localvar.COUNT_LIMIT}). {eventname.card_id}' {event.card_id} flagged to be limited.

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-3

Comment

Decline transaction and flag card to be limited if the number of transactions and total amount spent by card within the past 1 hour is above a set threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Velocity-3]
- Limit/Review Card
- Alert

Variables

```
CNT_LMT = cards__rules_thresholds["max_card_appr_count_1h"].threshold;

SUM_AMT_LMT = cards__rules_thresholds["max_card_sum_appr_amount_1h"].threshold;
SUM_AMT_LMT_CONV = (SUM_AMT_LMT * 1.0) / 100;

SUM_AMT_CONV = (event.appr_auth_cardid_sum_billamosr * 1.0) / 100;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.appr_auth_cardid_count_1h >= CNT_LMT
and
event.appr_auth_cardid_sum_billamosr >= SUM_AMT_LMT
```

Explanation

[Cards-Velocity-3] - Decline - Card transactions ({event.appr_auth_cardid_count_1h}) and spending ({localvar.SUM_AMT_CONV} {event.currency_code}) in the last 1h are over the limits ({localvar.CNT_LMT} and {localvar.SUM_AMT_LMT_CONV} {event.currency_code}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-4

Comment

Decline transaction if the number of transactions made by card within the last 3 minutes is above defined threshold.

Actions

- [Cards-Velocity-4]
- Alert

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_appr_count_3min"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.appr_auth_cardid_count_3m >= COUNT_LIMIT
```

Explanation

[Cards-Velocity-4] - Decline - Card transactions ({event.appr_auth_cardid_count_3m}) in the last 3 minutes is over the limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-5

Comment

Distance between current and previous ip location in the last hour exceeds 1000 km (1 hour)

Actions

- [Cards-Velocity-5]
- Limit/Review Card
- Alert

Variables

```
distance =
Location.distFromInKm(event.device_ip_latitude,event.device_ip_longitude,event.auth_cnp_cardid_p
rev_device_ip,event.auth_cnp_cardid_prev_device_sr) ?? 0.0;

DISTANCE_LIMIT = cards__rules_thresholds["max_distance_ip_location_km_1h"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
(event.auth_cnp_cardid_prev_device_ip ?? 0.0 != 0.0) and (event.auth_cnp_cardid_prev_device_sr ??
0.0 != 0.0)
and
distance >= DISTANCE_LIMIT
```

Explanation

[Cards-Velocity-5] - Decline - Distance between previous and current IP location in the last hour is higher than {localvar.DISTANCE_LIMIT} km.

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-6

Comment

Ratio of distinct card_id and distinct expiry dates per BIN and Merchant in the last 10m is below threshold, indicating a possible BIN attack.

Actions

- [Cards-Velocity-6]
- Alert

Variables

```
// curve3 (th3) is expected to have better results then the others (good compromise between
number of alerts and detection rate)
// other curves are still provided in case there is the need to adjust
binAttackThreshold = cards__rules_thresholds["acceptor_cardbin_curve_th"].threshold;
```

```
// acceptor_cardbin_cur = binary fields to split transactions above and below threshold curve
thresholdDecision = match
  if binAttackThreshold == 0 then return event.acceptorid_cardbin_cur0;
  else if binAttackThreshold == 1 then return event.acceptorid_cardbin_cur1;
  else if binAttackThreshold == 3 then return event.acceptorid_cardbin_cur3;
  else if binAttackThreshold == 5 then return event.acceptorid_cardbin_cur5;
  end
end
```

Condition

thresholdDecision

Explanation

[Cards-Velocity-6] - Decline - The ratio between distinct card_id ({event.acceptorid_cardbin_cntdist_csr}) and distinct exp_date ({event.acceptorid_cardbin_cntdist_car}) per BIN/Merchant 10m is below {localvar.binAttackThreshold}

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-7

Comment

Ratio of distinct card_id and distinct expiry dates per BIN and MCC in the last 10m is below threshold, indicating a possible BIN attack.

Actions

- [Cards-Velocity-7]
- Alert

Variables

```
// curve3 (th3) is expected to have better results then the others (good compromise between
number of alerts and detection rate)
// other curves are still provided in case there is the need to adjust
binAttackThreshold = cards__rules_thresholds["mcc_cardbin_curve_th"].threshold;

// acceptor_cardbin_cur = binary fields to split transactions above and below threshold curve
thresholdDecision = match
  if binAttackThreshold == 0 then return event.mcc_cardbin_cur0;
  else if binAttackThreshold == 1 then return event.mcc_cardbin_cur1;
  else if binAttackThreshold == 3 then return event.mcc_cardbin_cur3;
  else if binAttackThreshold == 5 then return event.mcc_cardbin_cur5;
  end
end
```

Condition

thresholdDecision

Explanation

[Cards-Velocity-7] - Decline - The ratio between distinct card_id ({event.mcc_cardbin_cntdist_cardid_10m}) and distinct exp_date ({event.mcc_cardbin_cntdist_cardexpdat}) per BIN/MCC 10m is below {localvar.binAttackThreshold}

Mark As

The outcome of this rule is: **decline**

Cards-CNP-1

Comment

Flag card to be limited if it has made multiple CNP transactions in a high number of unique merchants (acceptor_id) in the last 5 minutes. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-CNP-1]
- Limit/Review Card
- Alert

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_cnp_card_distinct_acceptor_5min"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.auth_cnp_cardid_countdist_acce >= COUNT_LIMIT
```

Explanation

[Cards-CNP-1] - Card CNP transactions in unique merchants ({event.auth_cnp_cardid_countdist_acce}) in the last 5 minutes are over the limit ({localvar.COUNT_LIMIT}). '{eventname.card_id}' {event.card_id} flagged to be limited.

Mark As

N/A

Cards-CNP-2

Comment

Decline and flag card to be limited if transaction is CNP and has high amount with an unusual currency. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-CNP-2]
- Limit/Review Card

- Alert

Variables

N/A

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '') and (event.card_presence_flag ?? false == false) and
(event.currency_code ?? '' != '')
and
// Limits
event.is_high_amount and
!common_currencies.contains(event.currency_code)
```

Explanation

[Cards-CNP-2] - Decline - Card CNP transaction with an high amount in an unusual currency.

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-1

Comment

Decline transaction and flag card to be limited if total amount spent by card within the past 24 hours at single Merchant (acceptor_id) is above a defined threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Threshold-1]
- Limit/Review Card
- Alert

Variables

```
SUM_AMOUNT_LIMIT = cards_rules_thresholds["max_card_acceptor_sum_appr_amount_24h"].threshold;
SUM_AMOUNT_LIMIT_CONV = (SUM_AMOUNT_LIMIT * 1.0) / 100;

SUM_AMOUNT_CONVERTED = (event.appr_auth_cardid_acceptor_sum_ * 1.0) / 100;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.appr_auth_cardid_acceptor_sum_ >= SUM_AMOUNT_LIMIT
```

Explanation

[Cards-Threshold-1] - Decline - Card spending ({localvar.SUM_AMOUNT_CONVERTED}{event.currency_code}) in the last 24 hours is above the limit ({localvar.SUM_AMOUNT_LIMIT_CONVERTED}{event.currency_code}). Card {event.card_id} flagged to be limited.

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-2

Comment

Decline transaction if number of transactions made by card within the past 4 weeks is above a set threshold.

Actions

- [Cards-Threshold-2]
- Alert

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_count_4w"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.auth_cardid_count_4w >= COUNT_LIMIT
```

Explanation

[Cards-Threshold-2] - Decline - Card transactions ({event.auth_cardid_count_4w}) in the last 4 weeks are above the limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-3

Comment

Decline transaction and flag card to be limited if amount spent by card within the past 4 weeks is above a set threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Threshold-3]
- Limit/Review Card
- Alert

Variables

```
SUM_AMOUNT_LIMIT = cards__rules_thresholds["max_card_sum_appr_amount_4w"].threshold;
SUM_AMOUNT_LIMIT_CONVERTED = (SUM_AMOUNT_LIMIT * 1.0) / 100;

SUM_AMOUNT_CONVERTED = (event.appr_auth_cardid_sum_billamoun * 1.0) / 100;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.appr_auth_cardid_sum_billamoun >= SUM_AMOUNT_LIMIT
```

Explanation

[Cards-Threshold-3] - Decline - Card spending ({localvar.SUM_AMOUNT_CONVERTED} {event.currency_code}) in the last 4 weeks is above the limit ({localvar.SUM_AMOUNT_LIMIT_CONVERTED} {event.currency_code}).

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-4

Comment

Decline and flag card to be limited if number of high amount transactions made by the same card within the past day is above a set threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Threshold-4]
- Limit/Review Card
- Alert

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_appr_count_high_amount_24h"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits (v1: high amount = 1000 USD, check is_high_amount)
event.high_amount_appr_auth_cardid_c >= COUNT_LIMIT
```

Explanation

[Cards-Threshold-4] - Decline - Card high amount transactions ({event.high_amount_appr_auth_cardid_c} above 1000.00 {event.currency_code}) in the last 24h are over the limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-5

Comment

Decline transaction and flag card to be limited if multiple transactions with a high cumulative amount made by the same card in the same merchant (acceptor_id) in the last 24h. Depends on 'List Management Service' to put the card in the configured list.

Actions

- Limit/Review Card
- Alert
- [Cards-Threshold-5]

Variables

```
CNT_LMT = cards__rules_thresholds["max_card_acceptor_appr_count_24h"].threshold;

SUM_AMNT_LMT = cards__rules_thresholds["max_card_acceptor_sum_appr_billamount_24h"].threshold;
SUM_A_LMT_CONV = (SUM_AMNT_LMT * 1.0) / 100;

SUM_A_CONV = (event.appr_auth_cardid_acceptor_sum_ * 1.0) / 100;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.appr_auth_cardid_acceptor_coun >= CNT_LMT and
event.appr_auth_cardid_acceptor_sum_ >= SUM_AMNT_LMT
```

Explanation

[Cards-Threshold-5] - Decline - Card transactions ({event.appr_auth_cardid_acceptor_coun}) and total amount ({localvar.SUM_A_CONV} {event.currency_code}) in the same merchant in the last 24h are over the limits ({localvar.CNT_LMT} and {localvar.SUM_A_LMT_CONV}).

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-6

Comment

Decline transaction and flag card to be limited if the number of distinct devices per card is higher than threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- Limit/Review Card
- Alert

- [Cards-Threshold-6]

Variables

```
DISTINCT_LIMIT = cards__rules_thresholds["max_cnp_card_distinct_device_24h"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
event.auth_cnp_distinct_device_id_pe >= DISTINCT_LIMIT
```

Explanation

[Cards-Threshold-6] - Decline - Distinct devices per card in the last 24 hours are above limit
({{localvar.DISTINCT_LIMIT}}). '{eventname.card_id}' {event.card_id} flagged to be limited.

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-1

Comment

Decline and flag card to be limited if the card has a high number of failed transactions in the last 24 hours.
Depends on 'List Management Service' to put the card in the configured list.

Actions

- Limit/Review Card
- Alert
- [Cards-Behavior-1]

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_declined_count_24h"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.decline_cardid_count_1d >= COUNT_LIMIT
```

Explanation

[Cards-Behavior-1] - Decline - Card failed transactions ({{event.decline_cardid_count_1d}}) in the last 24h
are over the limit ({{localvar.COUNT_LIMIT}}).

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-2

Comment

Flag card to be limited if the card had a same amount transaction in the same merchant (acceptor_id) in the last day.

Actions

- Limit/Review Card
- Alert
- [Cards-Behavior-2]

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_acceptor_amount_count_appr_24h"].threshold;
AMOUNT_LIMIT = cards__rules_thresholds["max_bill_amount"].threshold;

AMOUNT_CONVERTED = ((event.cardholder_bill_amount ?? 0)* 1.0) / 100;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '') and (event.cardholder_bill_amount ?? 0 != 0)
and
// Limits
event.appr_auth_cardid_acceptor_bill >= COUNT_LIMIT and
event.cardholder_bill_amount >= AMOUNT_LIMIT
```

Explanation

[Cards-Behavior-2] - Card transactions with the same amount in the same merchant ({event.appr_auth_cardid_acceptor_bill} of {localvar.AMOUNT_CONVERTED} {event.currency_code}) in the last 24h are over the limit ({localvar.COUNT_LIMIT}).

Mark As

N/A

Cards-Behavior-3

Comment

Decline transaction and flag card to be limited if this transaction is made on very high fraud rate acceptor, previously not used by a card within 6 months. Depends on 'List Management Service' to put the card in the configured list.

Actions

- Limit/Review Card
- Alert
- [Cards-Behavior-3]

Variables

```
FRAUD_RATE_LIMIT = cards__rules_thresholds["max_acceptor_fraud_rate_6M"].threshold;  
COUNT_LIMIT = cards__rules_thresholds["max_acceptor_count_appr_6M"].threshold;
```

Condition

```
// Lists  
event.card_id_not_blacklisted_nor_wh  
and  
// Filters  
(event.card_id ?? '' != '')  
// Limits  
and  
event.appr_auth_acceptorid_count_6m >= COUNT_LIMIT and  
event.fraud_ratio_acceptorid_6m > FRAUD_RATE_LIMIT and  
event.auth_cardid_acceptorid_count_6 == 0
```

Explanation

[Cards-Behavior-3] - Decline - Card transaction in an acceptor with fraud rate over the limit ({localvar.FRAUD_RATE_LIMIT}) and where the card had no history in the last 6 months.

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-4

Comment

Decline transaction if the device is infected with malware.

Actions

- [Cards-Behavior-4]
- Limit/Review Card
- Alert

Variables

N/A

Condition

```
// Lists  
event.card_id_not_blacklisted_nor_wh  
and  
// Filters  
(event.card_id ?? '' != '')  
and  
event.malware_indicator ?? false
```

Explanation

[Cards-Behavior-4] - Decline - Malware has been detected

Mark As

The outcome of this rule is: **decline**